

#DB3

02/12/25

Advanced SQL

select	projection list
from	relations to be used
where	selection/join conditions, nested queries
group by	grouping for aggregate functions
having	selection conditions for groups
order by	output order

Calculation sequence: from, where, group by, having, select, order by
(in what order the ~~query~~ ~~is~~ ~~executed~~ query is executed by the ~~the~~ program).

- Aggregate functions calculate ~~new values~~ ~~new values~~ for an entire column, e.g. sum of a column

COUNT(*) → count every tuple

COUNT(columnName) → count every non-null entry in the given column

- if an attribute in a query is in an aggregate function, then the remaining attributes must be in a "group by"?
- can't nest aggregate functions bc it is dependent on the group by (which can be only executed once per query)
- stacking can be done via subqueries though

~~Reading~~ aggregate functions

DB3

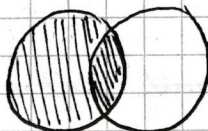
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Natural Join



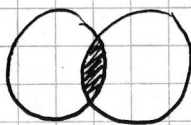
Animal Compound
→ get only tuples that define both an Animal and a Compound (since there can be empty Compounds)

Left Join

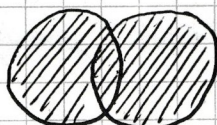


Animal Compound
→ get all Animals and if there are any, also info on their Compound

Inner Join



Full Outer



In exam: "You want to get all animals/customers/etc..."

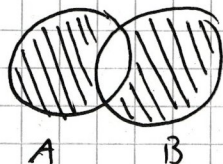
→ hint to use left join

Example

Animal	Compound
A	x
B	null
(Cross product)	
A	x
A	y
B	x
B	y
(inner join)	
Animal	Compound
A	x
(left join)	
Animal	"left" Compound
A	x
B	null
(full outer)	
Animal	"full" Compound
A	x
B	null
null	y

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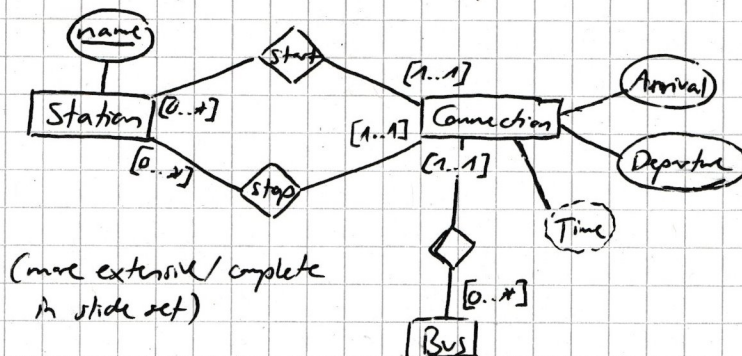
full outer without natural join by
defining where condition that includes
only tuples where given key is null

e.g.: where customer.id is null or order.id is null

Recursion via **named queries** and **with** notation expression



- creates a local temp table that can be queried to

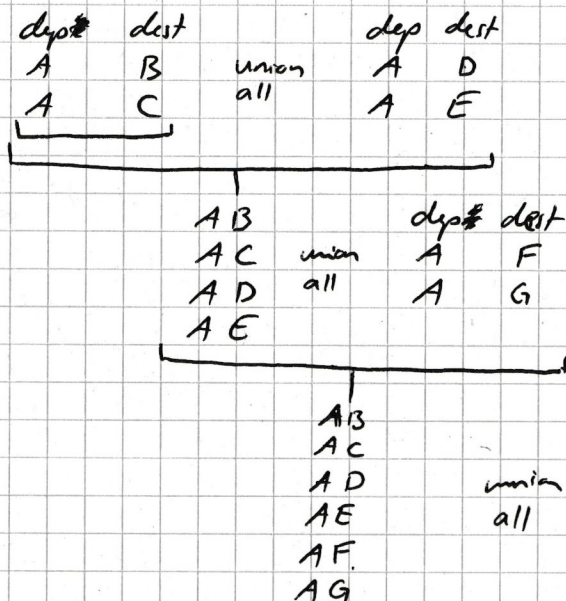


(more extensive/complete
in slide set)

with recursionName (attr1, attr2) as (
// init recursion
union
union all
// recursive step
)

num of attributes must match
num of attr in query (init, step)

← **all** keyword causes recursion, **stop** when the
latest table is empty and
does not carry any new information



union all: attributes must be same datatype, datatype is determined by
topmost first attribute defined in select.